



Original Article

Spontaneous quantity discrimination in the Australian sleepy lizard (*Tiliqua rugosa*)

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Animals employ quantitative abilities to gauge crucial aspects of their environment, such as food or predator density in a given area or the number of eggs in a nest. These quantitative skills hold ecological implications and can impact an animal's fitness. However, our comprehension of how these abilities intersect with environmental challenges remains limited. We tested for spontaneous quantity discrimination in the sleepy lizard (*Tiliqua rugosa*), a long-lived species known for forming life-long pair bonds in which pairs come together for the breeding season. We subjected lizards to a Y-maze experiment presenting both discrete (comparisons involving two vs. four, two vs. six, two vs. eight, four vs. six, four vs. eight, and six vs. eight pieces) and equivalent continuous combinations (single pieces differing in size) of pumpkin pieces. Our findings revealed that sleepy lizards exhibited spontaneous quantity discrimination, favoring the larger quantity in three discrete comparisons (eight vs. two, two vs. six, four vs. six) and two continuous comparisons (eight vs. two, eight vs. four). However, low accuracy may be attributable to difficulty in visual acuity exacerbated by difficulty discriminating quantities at small ratios. Furthermore, we found no evidence for a speed-accuracy trade-off in their decision-making process. Our results suggest that sleepy lizards likely employ multiple cues to estimate quantity accurately. Considering these results in the context of previous studies on various lizard species, it becomes apparent that lizards, as a group, offer a valuable model for unraveling the evolution of cognition, given their diverse ecology and sociobiology.

Key words: cognition, foraging, number sense, reptile, Squamata.

INTRODUCTION

Having at least a rudimentary ability to discriminate quantities can provide an adaptive advantage in the social as well as the non-social domain (Agrillo 2015; Agrillo and Bisazza 2017; Nieder 2020). For instance, the decision of whether to attack or retreat during an encounter may be based on differences in group size (e.g., McComb et al. 1994; Wilson et al. 2001; Tanner 2006; Benson-Amram et al. 2011) because each group's relative size can be related to the outcome of a conflict (Clifton 2020; Lanchester's laws; Lanchester 1916). Joining the larger of two groups can also be advantageous if bigger groups provide better predator protection through dilution (e.g., Hager and Helfman 1991; Kraus and Godin 1994; Agrillo and Bisazza 2017; Messina et al. 2022) or are more successful in deterring a predator by driving it away through alarm calls or mobbing (e.g., Coomes et al. 2019; Dutour et al. 2021). In terms of sexual selection, quantity discrimination can facilitate keeping track of the number of available mates and competitors in the environment and thereby influence mate choice or alternative reproductive tactics such as mate guarding (e.g., Carazo et al. 2009,

2012). The ability to discriminate quantities can also be beneficial while foraging or during habitat selection. For example, deciding whether to choose or leave a food patch and whether to defend it against competitors may be guided by patch size (e.g., Nelson and Jackson 2012; Agrillo 2015; Lucon-Xiccato et al. 2015; Agrillo and Bisazza 2017). Furthermore, knowledge about resource availability may guide future hunting decisions (Hunt et al. 2008; Rodríguez et al. 2015). Finally, habitat choice may be influenced by quantitative information if denser habitat provides more favorable conditions (Lucon-Xiccato et al. 2018). It is, therefore, not surprising that the ability to discriminate quantities has been demonstrated across a wide range of vertebrates and invertebrates (Agrillo 2015; Vallortigara 2017; Nieder 2020; Bortot et al. 2021).

Research investigating a species' natural ability to distinguish quantities in ecologically relevant contexts focuses on testing spontaneous quantity discrimination (e.g., Benson-Amram et al. 2011; Nelson and Jackson 2012; Agrillo 2015; Rodríguez et al. 2015; Agrillo and Bisazza 2017; Lucon-Xiccato et al. 2018). In order to do this, different quantities of relevant stimuli (conspecific cues, food, and other resources) are presented simultaneously, and an individual's response is recorded. On the one hand, if “going for

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more” is adaptive we expect that animals are, overall, more likely to respond to the larger quantity if they can detect the difference between the two options. On the other hand, if “going for less” is adaptive, we would expect a higher response rate toward the smaller quantity (Nieder 2020).

Response behavior often follows two patterns: the numerical distance effect and the numerical size effect (i.e., Weber’s law, Weber 1850). The less the distance between two quantities (the more similar they are), the harder they are to distinguish. Additionally, the difficulty of differentiating quantities also increases with increasing set size (larger quantities). The presence of these two effects suggests the approximate number system (estimation/discrimination of magnitudes based on their ratio) is used to judge quantities (Agrillo 2015; Nieder 2020). A second system, the object tracking system (tracking discrete entities in space; usually limited to up to 4–5 items), is characterized by the absence of a size and distance effect (Feigenson et al. 2004). Evidence for or against both systems is often inconclusive, however, especially if only small quantities (1–4 items) are tested because Weber’s law might only be reliably detected when large quantities (>5 items) are used. Some studies, however, have shown that performance does follow the numerical distance effect even when testing the discrimination of small quantities (e.g., Uller et al. 2003; Lucon-Xiccato et al. 2015; Gazzola et al. 2018; Szabo et al. 2021). This suggests that both systems might be used together to make judgments about quantity (Agrillo 2015; Nieder 2020).

To be able to choose the most beneficial quantity when multiple options are available, individuals need to process different information related to quantity, such as number, size, circumference, or area (Agrillo 2015; Vallortigara et al. 2022). When the options differ greatly, processing time is lower, and decisions can be made fast and accurately. If the options are very similar, then discrimination becomes harder. Consequently, accuracy to choose the most profitable option decreases, especially when time to choose is limited or, alternatively, when time is unlimited, accuracy can be maintained by increasing processing time (Chittka et al. 2009). This is called a speed-accuracy trade-off (SAT) and has been demonstrated across contexts such as foraging, mate selection, and predator avoidance (e.g., Chittka et al. 2003; Dyer and Chittka 2004; Ings and Chittka 2008; Hemingway et al. 2019; Jones et al. 2020). But, not all studies show the predicted speed-accuracy trade-off (e.g., Greis et al. 2022). When the task is easy (e.g., presenting only two options to choose from; Jones et al. 2020), there are no costs to making a suboptimal choice (e.g., Chittka et al. 2003) or when learning is prevented (e.g., Jones et al. 2020), SATs might not be detectable (Chittka et al. 2009).

Although studies on quantity discrimination are increasing across the whole animal kingdom, some groups, such as reptiles, are still underrepresented (Agrillo 2015) despite their great diversity (Pianka and Vitt 2003; Uetz et al. 2022). This dearth of data hampers our ability to uncover broad-scale evolutionary patterns. Nevertheless, three species of reptiles have been tested on their spontaneous quantity discrimination ability between different quantities of food. Italian wall lizards (*Podarcis sicula*) select larger dead fly larvae (*Musca domestica*) at ratios of 0.25, 0.5, 0.67, and 0.75 but are unable to discriminate different numbers of larvae (Miletto Petrazzini et al. 2017). Another lizard species, the gidgee skink (*Egernia stokesii*), is able to discriminate different numbers of single carrot pieces by choosing the larger number above chance at ratios including 0.25, 0.33, 0.5 (2 vs. 4) and 0.75 (but not 0.5–1 vs. 2 and 0.67). These

omnivorous skinks are, however, unable to select the larger quantity of differently sized carrot pieces (Szabo et al. 2021). Finally, Hermann’s tortoises (*Testudo hermanni*) select the larger quantity of tomato slices in both a test based on number and a test based on size at ratios 0.25, 0.5, 0.67, and 0.75 (Gazzola et al. 2018). What causes these marked differences in ability is still unclear and we are in dire need of more studies investigating the innate quantitative abilities of a broader range of reptile species to better understand why species differ in their spontaneous quantity discrimination ability.

Our primary aim was to test spontaneous quantity discrimination in the Australian sleepy lizard (*Tiliqua rugosa*), a species with an unusual social system in which males and females pair for life, are monogamous, but only come together during a 3-month breeding season (Bull 1987, 1988; Bull et al. 1998, 2017). Sleepy lizards are a large (26–31 cm snout-vent length), terrestrial, diurnal, long-lived (>50 years), omnivorous skink species from Australia with live birth (Bull 1987; Dubas and Bull 1991; Cogger 2014). They are an ideal species to test food-based spontaneous quantity discrimination because they are active foragers that visually search for food and are also able to discriminate both insect and plant chemicals (Cooper 2000; Zuri and Bull 2000; Wohlfeil 2008). Previously, they have also demonstrated the ability to associate a visual cue with a food reward and reverse this association successfully (Szabo and Whiting 2020) but possess poor inhibitory control, repeatedly trying to access food through a mesh barrier instead of detouring to the side (Szabo et al. 2020). We based our methodology on previous studies but improved upon previous testing procedures (Miletto Petrazzini et al. 2017; Gazzola et al. 2018; Szabo et al. 2021) by presenting both size and number-based discriminations within the same day. This allowed us to examine within-individual differences in quantitative abilities based on continuous and discrete information. Finally, we also investigated speed-accuracy trade-offs to test whether individuals sacrificed accuracy for speed during their decision-making process.

METHODS

Animals and husbandry

We used 10 adult female sleepy lizards (*T. rugosa*) in this study. These lizards were previously caught from multiple locations in New South Wales and had been housed in the research facility for 16 months prior to the commencement of this study. Lizards were identified by individual markings and enclosure number. We housed lizards individually in open-topped plastic tubs measuring 800 (L) × 600 (W) × 450 (H) mm in a temperature-controlled room. Room temperature was maintained between 25 and 26 °C, and ambient humidity ranged between 38% and 60%. One end of each tub was fitted with a strong UV-bulb and heat lamp, designed for desert-living species, on a 12 h light cycle (7:00 AM to 7:00 PM). Enclosures contained a single refuge as well as sticks, bark, rocks, and leaves for enrichment. Each enclosure was lined with butcher’s paper, which was replaced when soiled.

Outside of the testing period, we fed lizards three times per week (with a salad mix adding one source of protein such as dog food or boiled eggs once a week) on Mondays, Wednesdays, and Fridays between 07:00 and 15:00 h. After each trial, lizards received a reduced volume of food to ensure motivation to participate in trials was maintained while ensuring their nutritional needs were met. Water was provided once a week on Thursdays.

Set up and maze

We divided the lizards into two groups of five (A and B). As female *T. rugosa* tend to forage in the afternoons after basking during the morning (Kerr and Bull 2006), testing took place between 11:30 and 15:30 h to help ensure lizards would be motivated to eat. We tested group A on Mondays, Tuesdays, Wednesdays, and Thursdays between 26 June and 27 September 2021. Group B was tested on Wednesdays, Thursdays, Fridays, and Saturdays between 6 April and 13 May 2022.

A purpose-built Y-maze was constructed from 6 mm black acrylic (Figure 1a). The maze had three arms of equal length arranged at 120 degrees around a triangular central chamber. Each

arm had a strip of acrylic mounted to each wall, approximately 2.5 cm from the start of the arm, to act as markers (Figure 1a). To reduce the impacts of handling stress from resetting the lizard to a starting arm, each arm could be closed off with a panel and became the start arm for the subsequent trial (see Supplementary Video M1). The maze design was based on standard Y-mazes used in cognition and choice testing with rodents, scaled to an appropriate size for sleepy lizards (Parker et al. 2021). The arms each measured 428 (L) × 322 (W) × 400 mm (H) (Figure 1a). The maze was placed on the floor approximately 3 m from the lizards' home enclosures, and a GoPro HERO7 Silver camera (set to 1080 p and 30 fps) was mounted directly above the maze to record all trials.

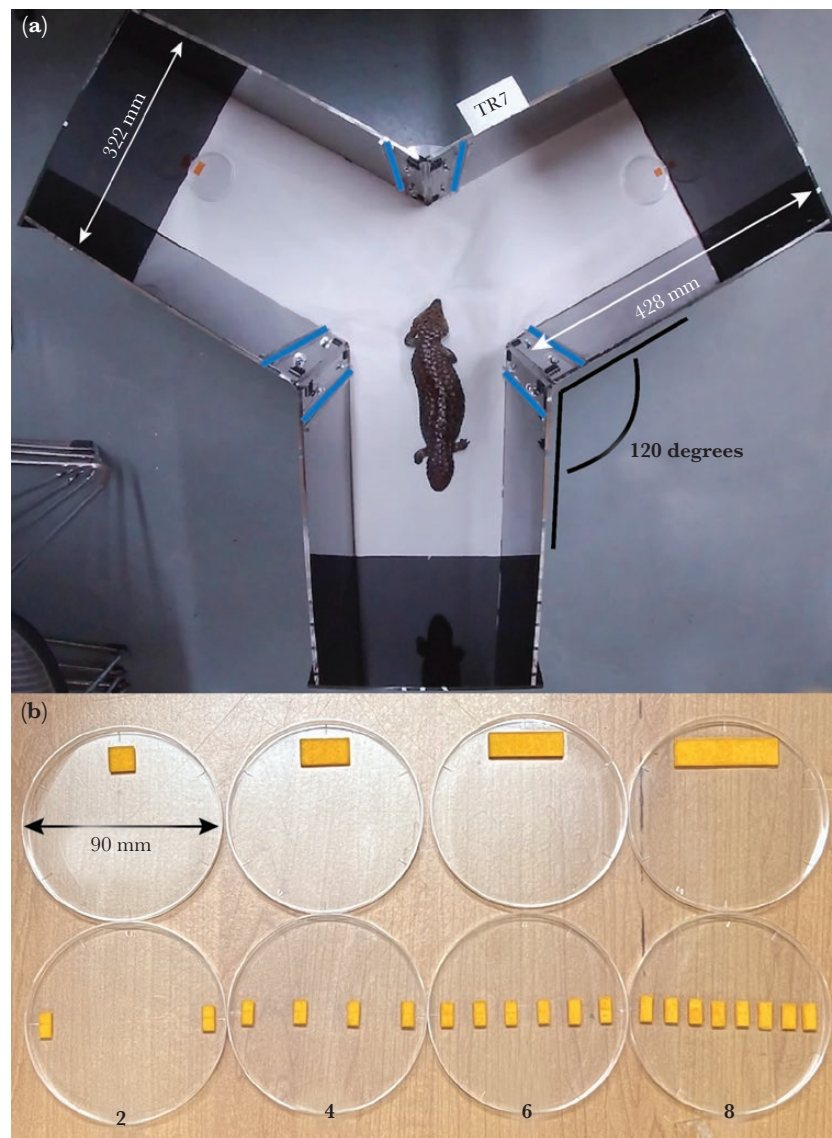


Figure 1

Picture of the Y-maze used to test spontaneous quantity discrimination (a) and a picture of the used pumpkin stimuli (b). (A) The maze was built from 6 mm black acrylic. Arms were 428 mm in length, 322 mm in width and the maze was 400 mm high. Arms were arranged at 120 degrees around a triangular central chamber. A strip of acrylic 2.5 cm from the start of an arm was used as a marker (blue lines). During trials, stimuli were placed at the end of two arms. (b) All pumpkin pieces were prepared fresh each day. We used a mandolin to slice the pieces into a consistent thickness of 3 mm, then pressed the slices into a 6 mm grid and cut the required size with a sharp knife. Pieces in the discrete test (number discrimination) were 6 × 12 × 3 mm in size and we used 2, 4, 6, or 8 pieces. In the continuous test (size discrimination), we used single pieces of 12, 24, 36, and 48 mm length (12 (W) × 3 (H) mm). All stimuli were presented in closed, transparent Petri dishes of 90 mm diameter.

To reduce the influence of olfactory cues on decisions, we placed pieces of pumpkin along the outside perimeter of the maze in order to saturate the maze with the smell of pumpkin.

Food preference test and stimulus selection

We first determined the most suitable food to use as a test stimulus by conducting a simple food preference test on each lizard prior to the study. Lizards were tested in the afternoon, after feeding, to ensure choice was motivated by preference and not hunger alone. Lizards were offered an equal-sized piece of butternut squash, yellow capsicum, red capsicum, strawberry, red cabbage, or carrot, placed equal distances apart and equidistant from the lizard. We recorded the order in which these foods were consumed for each lizard, and the test was repeated three times. All lizards ate butternut squash, yellow capsicum, and carrot before any of the other foods in each test (Supplementary Table S1). We then offered all the lizards two of these three foods at a time and recorded which food was consumed first. We repeated this until each pair of food choices (squash vs. capsicum, squash vs. carrot, carrot vs. capsicum) had been presented a total of three times. Nine lizards ate butternut squash first when available in all tests; however, one lizard preferred yellow capsicum to butternut squash but preferred butternut squash to carrot (Supplementary Table S2). As butternut squash was still a preferred food item for this lizard and was consumed in all tests, we selected butternut squash as the reward and stimulus for all lizards throughout testing.

We used the same pieces of butternut squash stimuli within a testing day and prepared fresh pieces each day. The squash was sliced using a mandolin to ensure a consistent thickness of 3 mm. We lightly pressed the slices into a 6 mm grid and then cut the required pieces with a sharp knife to ensure consistency in size (Figure 1b).

General procedure

For all testing, we gently carried lizards from their enclosure and placed them into the central chamber of the maze. The floor of the maze was lined with paper that was changed between each lizard to prevent conspecific scent cues from influencing choices. Each lizard was given at least 5 min in the maze to recover from handling and to acclimate to the maze. During this time, all doors remained open, no stimuli were present, and the lizard was free to explore the maze (see Supplementary Video M1). After 5 min had passed, the door to the arm the lizard was currently in was closed, and the trial started (see Supplementary Video M1). If the lizard was not in an arm at the end of the acclimation period, it was given additional time to move into an arm, which was then closed behind it. At no point after the initial placement were the lizards handled again until the trials were completed, thereby keeping handling stress to a minimum (Langkilde and Shine 2006).

We placed a covered transparent Petri dish (90 mm diameter; Figure 1b) containing a stimulus (required amount of butternut pumpkin) in the center-back of each of the remaining arms. We chose to use Petri dishes, as they allowed us to enclose the stimuli and provide unobstructed visual access. A single piece of pumpkin ($6 \times 12 \times 3$ mm) was placed in the center of the maze (triangular compartment) while the start-arm door was closed. This piece of pumpkin served both to align the head of the lizard equidistant and centrally between the two options and to prevent the lizards from losing motivation throughout testing because they were not allowed to consume the stimuli. Lizards consumed the center piece in 1363

of 1388 trials (in 25 trials, the pieces were not eaten). The door was then opened, the researcher stepped away from the maze and out of view of the lizard, and the lizard was able to consume the central pumpkin and then make a choice between the two stimuli (Petri dishes; see Supplementary Video M1). Lizards were considered to have made a choice once both of their front legs passed the markers (2.5 cm from the start of the arm) on the walls of the arm. Once a choice was made and the lizard had moved far enough into an arm, the door to that arm was closed with the lizard inside, and the Petri dishes were removed from the maze arms (see Supplementary Video M1). The arm the lizard occupied then became the start arm for the subsequent trial. This allowed for the maze to be “reset” without needing to handle the lizard.

Side preference test

We tested for side preference on two separate days prior to any quantity discrimination testing. Two Petri dishes (90 mm diameter) were placed in the center at the end of either arm that the lizard was not occupying, each containing a single piece of butternut pumpkin ($12 \times 12 \times 3$ mm). Pumpkin was only placed in the center of the maze for the first two or three trials, at which point they were sufficiently motivated to walk into the center of the maze. The trial began when the door was opened and ran until the lizard chose a Petri dish. Each lizard received 13 trials conducted within a single day with an inter-trial interval of at least 30 s.

Once the lizard had chosen a dish and moved into the corresponding arm of the maze, we closed the door to that arm, recorded their choice, and allowed the lizard to eat the pumpkin within the chosen dish. The non-selected dish was removed from the maze, and the pumpkin was replaced with a fresh piece. Once the lizard had finished eating the pumpkin it chose, the empty dish was removed, wiped with 70% ethanol, and left outside and away from the maze to dry. We used a new dish with a fresh piece of pumpkin for the following trial so that one dish was always empty and drying outside of the maze during any given trial. This ensured that the alcohol had completely evaporated. The arm the lizard was in then became the start arm for the next trial, following the reset procedure described above.

Quantity discrimination test

We presented lizards with six combinations of number or size quantities: two versus four, two versus six, two versus eight, four versus six, four versus eight, and six versus eight (ratio: 0.5, 0.33, 0.25, 0.67, 0.5, and 0.75, respectively; Figure 1b) each day, thus, each lizard received 12 trials per day with an inter-trial interval of at least 30 s. In the number-based test, we presented multiple pieces of the same size ($6 \times 12 \times 3$ mm, discrete test), and in the size-based test, we presented single pieces of different sizes (12, 24, 36, $48 \times 12 \times 3$ mm, continuous test). Each combination was presented to each lizard only once within a day. We randomized the order of presentation across all 12 trials for each day, and the order the lizards were tested within a day was randomized for each day. The side the larger quantity (left or right) was placed was pseudo-randomized so that the larger quantity was never placed on the same side more than twice in a row. Each trial began when the door was opened and ended when the lizard chose a stimulus or failed the trial. Once a choice was made, the door to the arm was closed, both Petri dishes were removed from the maze, and the maze was reset following the previously described procedure (see Supplementary Video M1).

We discarded any trials in which the lizard had not started moving toward the center within 3 min of the door opening or trials in which the lizard did not start moving toward a choice arm (either arm containing a Petri dish with pumpkin) within 3 min of consuming the central pumpkin. When a lizard failed a trial, it was immediately removed from the apparatus and returned to its enclosure. Lizards that failed a trial were not tested again until the next testing day. Trials that were missed after a lizard failed were repeated in the days following the original 15 test days (due to SD card corruption, data from one test day for TR6 were not recorded), receiving a new trial number above 15 and recording trials within the new session starting with trial 1.

Video scoring

The chosen quantity was recorded immediately after the lizard had made a choice in each trial. One of us (M.H.) also recorded latency—the amount of time taken from consuming the pumpkin in the central chamber to making a choice, from the recorded footage. M.H. was blind to both the stimuli and test day. Due to the unique markings on each lizard, it was not possible to blind M.H. to the individual lizard's identity.

Ethical statement

We followed the guidelines provided by the Association for the Study of Animal Behaviour/Animal Behaviour Society for the treatment of animals in behavioral research and teaching (ASAB and ABS 2022). The experiment included noninvasive behavioral observations approved by the Macquarie University Animal Ethics Committee (ARA # 2013/031). The collection of lizards was approved by the New South Wales National Parks and Wildlife Service, Office of Environment and Heritage (License # SL101972). Animals were captured by hand or Elliott traps and transported back to Macquarie University, NSW, by car in cloth bags a maximum of 1 week after capture.

Statistical analyses

Analyses were run in R version 4.0.3 (R Core Team 2020). We report our results as suggested by Muff and colleagues (2022): $P > 0.1$ no evidence, $0.1 < P < 0.05$ weak evidence, $0.05 < P < 0.01$ moderate evidence, $0.01 < P < 0.001$ strong evidence, $P < 0.001$ very strong evidence. Two of the 10 initial test lizards were excluded because they did not engage with the task, and all analyses were performed on the data from the remaining eight females only.

Side preference test

The side preference test investigated if lizards innately preferred one side over the other. We used a binomial test to analyze if the number of choices to the right side was greater than expected by chance.

Accuracy

To test if lizards were more likely to choose the larger quantity of food, we used a Binomial Bayesian generalized linear mixed model (GLMM, package MCMCglmm; Hadfield 2010) with the larger/smaller quantity chosen (larger = 1 and smaller = 0, Bernoulli variable) as the response variable. To investigate differences across test types (continuous vs. discrete; categorical) and across comparisons (8 vs. 2, 2 vs. 6, 8 vs. 4, 2 vs. 4, 4 vs. 6, 6 vs. 8; categorical), we included both and their interaction as fixed effects. We included test group (group A and B; categorical) and body condition (scaled mass index—SMI; continuous; Peig and Green 2009) as fixed effects

to determine if either one had an influence on choice accuracy. The model included a random intercept of animal identity and a random slope of trial nested in day (session) to account for repeated measures and autocorrelation between successive choices. In all models, “trials” and “day” were included as continuous variables. To investigate if accuracy was above chance across comparisons in each test, we applied post hoc least squares means tests (LSM, package emmeans; Lenth 2021). We also compared accuracy between tests within each comparison using post hoc LSM tests. We assumed statistical significance ($P < 0.05$) if the 95% confidence interval (CI) did not, or barely (< 0.01) crossed 0.

Speed-accuracy trade-off

We wanted to know if the time taken to make a choice was different for correct (larger quantity) and incorrect (smaller quantity) choices. To this end, we used a Gaussian Bayesian GLMM (Hadfield 2010) with the log-transformed latency as the response variable (treatment pooled). We log-transformed the response variable to ensure a better fit to a Gaussian distribution. As fixed effects, we included (1) accuracy (larger = 1 and smaller = 0) in interaction with comparison (8 vs. 2, 2 vs. 6, 8 vs. 4, 2 vs. 4, 4 vs. 6, 6 vs. 8) to investigate the effect of discrimination difficulty; (2) test type (discrete or continuous) in interaction with comparison to investigate differences in latency across tests but within each comparison; (3) test group (A or B); (4) day, trial and start time (time point the lizard started a session each testing day, this was randomized each day) to look at changes in latency within a day and across the whole experiment; and (5) body condition (SMI). Day, trial, start time, and body condition were included as continuous variables and as measures of motivation. The results of interactions were then investigated using post hoc LSM tests.

To investigate if our lizards showed a speed-accuracy trade-off (SAT) and if within- and between-individual differences in SAT occurred, we used within-subject centering (van de Pol and Wright 2009). Within-individual variation in accuracy is related to phenotypic plasticity in decision time, while between-individual variation might also include effects of personality or the experiment. Within-subject centering allows to separate these effects (Moiron et al 2016). We attained a measure of between-subject variation by calculating the mean latency ($\bar{x}_j$) across and within comparisons and calculated within-subject variation by subtracting the mean latency from the individual latency ($x_{ij} - \bar{x}_j$) both across and within comparisons. We analyzed SAT and its variation using Binomial Bayesian GLMMs with the accuracy (larger = 1 and smaller = 0) as the response variable. First, we built a model to investigate within- and between-subject variation separately by including the mean ($\bar{x}_j$) and ($x_{ij} - \bar{x}_j$) as fixed effects, respectively, both in interaction with test type (continuous versus discrete). Additionally, we included trial, day, and time of day (measures of motivation; continuous) as fixed effects. Before analysis, time of day was converted into a numeric variable with low values reflecting the earliest time any lizard was tested, while large numbers reflected the latest time. Second, we built another model to investigate the correlation of accuracy and latency (i.e., SAT) and the difference of within- and between-subject variation in SAT by including individual latency (x_{ij}) in interaction with test and mean latency ($\bar{x}_j$) as fixed effects. Two separate models were needed as within-subject variation was derived by subtracting the mean latency from the individual latency. Afterwards, we investigated the results of the interactions in both models using post hoc tests for estimated marginal means of linear trends (EMM, package emmeans; Lenth 2021).



Box plots of performance accuracy in both the continuous (blue) and the discrete (yellow) test separated across combinations. The bold line represents the median, the upper and lower edge of the box the upper and lower quartile, the top and bottom edge of the whisker the maximum and minimum (outliers are not shown). * Significant difference according to the credible interval not or barely (<0.01) crossing 0. Single * indicates significant difference from chance. Lines connecting boxes indicate significant differences across combinations. The dotted line indicates chance level. For a graphical representation of individual data, see [Supplementary Figure S1](#).

individual consistency in accuracy (Binomial model, results reported on the link scale) and latency (log-transformed, Gaussian model) measures.

Side preference test

All models that we ran to investigate SAT included a random intercept of animal identity to account for repeated measures and autocorrelation between successive choices. For all Bayesian models, we made sure that no autocorrelation occurred (correlation between lags < 0.1 ; [Hadfield 2010](#)), that the MCMC chain sufficiently mixed (by visually inspecting plots; [Hadfield 2010](#)) and was run long enough (by passing the Heidelberg and Welch diagnostic tests; [Hadfield 2010](#)).

Repeatability

Lizards chose the larger quantity above chance when choosing between 8 versus 2 (LSM, marginal mean = 0.561, CI_{low} = 0.062, CI_{up} = 1.078) and 8 versus 4 (LSM, marginal mean = 0.524, CI_{low} = 0.032, CI_{up} = 1.045; [Supplementary Table S5; Figure 2](#)) in the continuous test. In the discrete test, lizards chose the larger quantity more often in comparisons 8 versus 2 (LSM, marginal mean = 0.551, CI_{low} = 0.032, CI_{up} = 1.048), 2 versus 6 (LSM, marginal mean = 0.550, CI_{low} = 0.053, CI_{up} = 1.066) and 4 versus 6 (LSM, marginal mean = 0.708, CI_{low} = 0.200, CI_{up} = 1.220; [Supplementary Table S5; Figure 2](#)). Overall, when comparing

accuracy between the continuous and discrete test within comparisons, we found no evidence that lizards' accuracy differed between the test types (Supplementary Table S5). However, lizards were more accurate in the discrete test when choosing 4 versus 6 (LSM, estimate = 0.795, CI_{low} = 0.165, CI_{up} = 1.431), 2 versus 6 (LSM, estimate = 0.639, CI_{low} = 0.010, CI_{up} = 1.279) and marginally more accurate when choosing 8 versus 2 (LSM, estimate = 0.637, CI_{low} = -0.009, CI_{up} = 1.269), compared to 6 versus 8 (Supplementary Table S5; Figure 2). We found no evidence that test group (GLMM, estimate = 0.166, CI_{low} = -0.357, CI_{up} = 0.621, P = 0.970; Supplementary Table S4) or body condition (GLMM, estimate = 0.001, CI_{low} = -0.003, CI_{up} = 0.005, P = 0.536; Supplementary Table S4) were associated with accuracy.

Speed-accuracy trade-off

We found no evidence that latency to choice differed between test groups (GLMM, estimate = 0.244, CI_{low} = -0.231, CI_{up} = 0.709, P = 0.243; Supplementary Table S6) or that latency was associated with body condition (GLMM, estimate = -0.001, CI_{low} = -0.004, CI_{up} = 0.003, P = 0.860; Supplementary Table S6). We found very strong evidence that latency increased across days (GLMM, estimate = 0.021, CI_{low} = 0.015, CI_{up} = 0.027, P = 0.00001; Supplementary Table S6) but decreased across trials (GLMM, estimate = -0.023, CI_{low} = -0.029, CI_{up} = -0.016, P = 0.00001; Supplementary Table S6). Furthermore, we found weak evidence that latency decreased with start time (GLMM, estimate = -0.001, CI_{low} = -0.002, CI_{up} = 0.001, P = 0.076; Supplementary Table S6). We found no evidence that latency differed across accurate and inaccurate choices within each comparison (Supplementary Table S7) or that latency differed across test types within comparisons (Supplementary Table S7).

We found no evidence for within- or between-individual variation in SAT when comparisons were pooled (within-individual variation: $EMM_{continuous}$, marginal mean = 0.007, CI_{low} = -0.011, CI_{up} = 0.024; $EMM_{discrete}$, marginal mean = 0.002, CI_{low} = -0.019, CI_{up} = 0.021; between-individual variation: $EMM_{continuous}$, marginal mean = 0.005, CI_{low} = -0.071, CI_{up} = 0.087; $EMM_{discrete}$, marginal mean = -0.007, CI_{low} = -0.082, CI_{up} = 0.071; Supplementary Tables S8 and S9) nor when we looked at each comparison separately (Supplementary Tables S10–S15). We also found no evidence that latency was correlated with accuracy ($EMM_{continuous}$, marginal mean = 0.003, CI_{low} = -0.014, CI_{up} = 0.019; $EMM_{discrete}$, marginal mean = -0.006, CI_{low} = -0.028, CI_{up} = 0.013; Supplementary Table S16; Figure 3) or for a difference between within- and between-individual differences (GLMM, estimate = 0.012, CI_{low} = -0.052, CI_{up} = 0.078, P = 0.709; Supplementary Table S16). We did, however, find moderate evidence that accuracy increased across trials within a day (GLMM, estimate = 0.039, CI_{low} = 0.001, CI_{up} = 0.078, P = 0.044; Supplementary Table S8) and found very strong evidence that accuracy decreased across days (GLMM, estimate = -0.074, CI_{low} = -0.107, CI_{up} = -0.041, P = 0.00002; Supplementary Table S8) and increased with time of day (GLMM, estimate = 0.008, CI_{low} = 0.002, CI_{up} = 0.015, P = 0.007; Supplementary Table S8).

Repeatability

When looking at choice accuracy and choice latency separately (contrary to SAT when both were considered together), we found no evidence for individual repeatability in choice accuracy (R = 0.006, CI_{low} = 0, CI_{up} = 0.019) but found evidence for

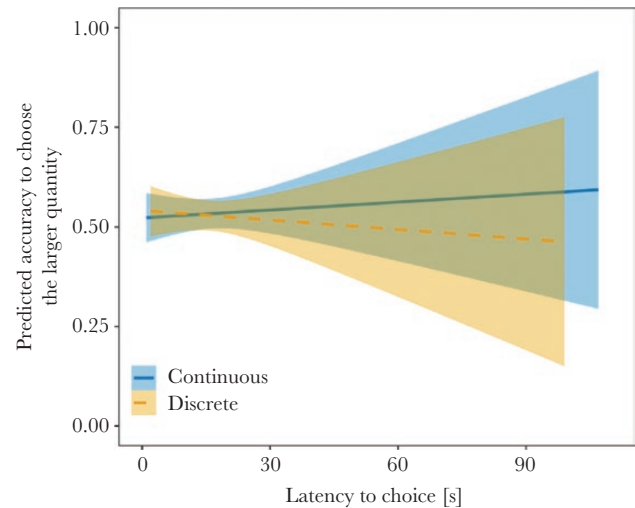


Figure 3

Correlation between predicted accuracy and latency for both the continuous test (blue, solid line) and the discrete test (yellow, dashed line). The shaded area represents the 95% confidence interval.

individual repeatability of choice latency (R = 0.213, CI_{low} = 0.058, CI_{up} = 0.388).

DISCUSSION

Overall, our lizards selected the larger quantity above chance when choosing between individual food items in 8 versus 2, 2 versus 6, and 4 versus 6. While they chose single food items differing in size above chance in 8 versus 4 and 2 versus 6. Accuracy (choosing the larger quantity) increased across trials within a day but decreased across days, and it was not repeatable. We found no evidence for a speed-accuracy trade-off and within- or between-individual differences in SAT. Latency was, however, strongly related to day and trial, albeit in different directions, and it was moderately repeatable.

Lizards were more accurate when both continuous and discrete information (number-based discrimination) was available compared to when only continuous information was provided (size-based discrimination). This is similar to what has been documented in gidgee skinks (Szabo et al. 2021) and in mosquitofish; the latter were better at discriminating two from three when both discrete and continuous information was available rather than only discrete or continuous information (Agrillo et al. 2011). Contrary to our previous study on gidgee skinks, sleepy lizards showed some ability to solely rely on continuous information at least when the discriminations were easy (quantities with larger difference). Lizards chose above chance in the continuous discrimination between 8 versus 2 and 8 versus 4, while they failed at the discrete discriminations 8 versus 4 and 2 versus 4 and performed very poorly in 6 versus 8. These inconsistencies could be partly explained by issues caused by Weber's law (Weber 1850) and with visual acuity (New and Bull 2011). At least in the continuous test, lizards only reliably discriminated single pumpkin pieces differing in size when the difference between them was rather large. Issues with visual acuity might have enhanced this effect. Low visual acuity might have also contributed to low accuracy in the discrete test. Single food items were placed across the whole diameter of the Petri dish (Figure 1b). The close proximity of each piece may have caused them to be perceived as one single

strip, especially in trials presenting six or eight pieces, making them indistinguishable. A different design such as using a vertical board presenting large single pieces spread out over a larger area, similar to what was used in a trained quantity discrimination test in Asian freshwater turtles (Lin et al. 2021), may be more effective in this species. Furthermore, such an approach would allow for the control of continuous information to give a fuller understanding of what information lizards rely on when judging quantities (e.g., Potrich et al. 2022; Zanon et al. 2022).

Our methodology was designed to avoid learning related to quantity by removing the chosen quantity rather than letting lizards consume it (see Miletto Petrazzini et al. 2017; Gazzola et al. 2018; Szabo et al. 2021). Although this did not seem to impact decisions in previous studies (Szabo et al. 2021; Greis et al. 2022), it seems to have had an impact on accuracy, latency, and motivation in the current study. First, our analyses demonstrate a reduction of accuracy across days while it improved within a day (trials). Removal of the chosen stimulus might have acted as negative punishment, possibly causing lizards to choose arbitrarily by the end of the experiment. Latency was also related to trial and day, although in the opposite direction compared to accuracy—decreasing across trials but increasing across days. Over the course of the experiment, lizards' choice behavior changed. Initially, in the experiment, they would directly approach their chosen quantity after reaching the branching point of the maze; however, during subsequent trials, they increasingly turned around and went to the other arm, thereby increasing latency. Furthermore, the decrease in latency across trials also indicates a change in behavior within a day. If lizards were indeed negatively punished by the removal of the chosen quantity, we might expect them to increase time to choose such quantities in subsequent trials, which matches our findings, rather than choosing arbitrarily which would likely decrease latency. Together, this suggests that “frustration,” and not trial fatigue, may have influenced their choice behavior. Finally, we can rule out that satiation caused changes in behavior across trials because the amount fed during each test day made up only a fraction (ca. 1.3%) of their regular food portion, and lizards were tested before being fed on any given day. Additionally, lizards consumed the center squash piece in 1363 of 1388 (98.2%) trials.

We found no evidence of a speed-accuracy trade-off either at the group level (all animals combined) or at the individual level and no difference in latency between accurate and inaccurate choices nor between continuous and discrete tests. Our previous work (Greis et al. 2022) and a study of poison frogs (Khatiwada and Burmeister, 2022) showed similar results, although gidgee skinks demonstrated a speed-accuracy alignment when data from all test individuals were considered. Our design and the low task difficulty might have prevented the formation of SATs across studies. First, there was no cost to choosing the smaller quantity or making no choice at all. Second, giving only two choices may not pose enough challenge for animals to trade-off speed and accuracy (e.g., Chittka et al. 2009; Jones et al. 2020). As researchers, we may be underestimating how difficult a task really is for animals. To really understand when animals need to make a trade-off between the speed of a decision and the accuracy of a choice, we need to design more challenging tasks that include relevant costs and risks (e.g., Dyer and Chittka 2004; Ings and Chittka 2008).

Sleepy lizard accuracy was not repeatable ($R = 0.006$), while latency was repeatable ($R = 0.213$), when accounting for both comparison (i.e., 8 vs. 2, 2 vs. 6, 8 vs. 4, 2 vs. 4, 4 vs. 6, 6 vs. 8) and test type (discrete and continuous), albeit at a low level. These results

support our previous finding in gidgee skinks, which lacked individual repeatability in accuracy ($0.1 < R < 0.2$; Szabo et al. 2021) but were repeatable in latency ($R = 0.228\text{--}0.493$; Greis et al. 2022). It is possible that without being able to learn about quantity by consuming the chosen stimuli as a reward, accuracy measures might be too variable or dependent on too many non-cognitive factors (e.g., day-to-day changes in motivation increasing within-individual variation across choices) to be repeatable across individuals as high within-individual variation decreases repeatability. Conversely, latency might be more dependent on lizard size (independent of body condition), leg length, or leg strength, rather than motivation. Therefore, individuals are consistent across trials, but depending on their size, leg length, or leg strength they differ from each other. On average, studies on behavior report individual repeatability of $R = 0.37$, and our repeatability measures of latency fall within the reported range ($0.22 \leq R \leq 0.27$) for ectotherms (Bell et al. 2009).

Among lizards, sleepy lizards were more similar in choice accuracy to gidgee skinks (Szabo et al. 2021) than Italian wall lizards (Miletto Petrazzini et al. 2017). Both diet as well as sociality might influence numerical competency across species. For example, species that forage on single prey items might not need sophisticated numerical abilities. Similarly, less social species might have less need for counting abilities. However, in general, animals have cognitive access to quantities in their environment and should possess basic quantitative abilities (Giaquinto 2018; Kobylkov et al. 2022). In order to understand the ecological and social factors that govern quantity discrimination, we need to test many species. As a precursor, it is still worth examining commonalities between a limited number of species. In terms of diet, both gidgee skinks and sleepy lizards are omnivorous (Cogger 2014), while wall lizards are insectivorous (Capula et al. 1993; Rugiero 1994). Some of the variance in performance may, therefore, be explained by diet. However, sleepy lizards chose above chance in some continuous discrimination tasks where gidgee skinks failed. In terms of sociality, gidgee skinks and sleepy lizards are known for forming stable pair bonds, although the nature of their pair bonds differ (gidgees remain in stable family groups, sleepy lizards pair up for 3 months in breeding season; Whiting and While 2017). On the other hand, wall lizards do not form pair bonds, although being territorial, they may track individuals in their neighborhood. Sociality can, therefore, not be ruled out completely. Importantly, sample sizes for both gidgee skinks and sleepy lizards were moderate, and therefore, these results should be viewed simply as a starting point. Overall, we still have too few studies investigating spontaneous quantity discrimination across lizard species to really understand the factors that influence quantitative abilities. Nevertheless, given the variation we are finding, lizards may be a good model for understanding the mechanisms and factors governing quantity discrimination more broadly.

SUPPLEMENTARY MATERIAL

Supplementary material can be found at <http://www.behco.oxfordjournals.org/>

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AUTHOR CONTRIBUTIONS

Birgit Szabo (Conceptualization [Equal], Formal analysis [Lead], Methodology [Equal], Validation [Equal], Visualization [Equal], Writing – original draft [Lead], Writing – review & editing [Equal]), Madeleine Holmes (Data curation [Equal], Investigation [Lead], Methodology [Equal], Writing – original draft [Supporting], Writing – review & editing [Equal]), Benjamin J. Ashton (Conceptualization [Equal], Funding acquisition [Equal], Methodology [Equal], Project administration [Equal], Supervision [Equal], Writing – original draft [Equal], Writing – review & editing [Equal]), and Martin J. Whiting (Conceptualization [Equal], Funding acquisition [Equal], Methodology [Equal], Project administration [Equal], Resources [Equal], Supervision [Equal], Writing – original draft [Equal], Writing – review & editing [Equal])

DATA AVAILABILITY

Analyses reported in this article can be reproduced using the data provided by Szabo et al. (2023a, b).

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